

Rio S. Espinosa

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PROJECTS

ProMorph: Melanoma Skin Lesion Classification Model with Proportional Dilation — [Python](#), [Tensorflow](#), [ML](#), [Keras](#), [Scikit](#)

- Developed a binary classification model to distinguish malignant and benign skin lesions, aiding early and non-invasive skin cancer detection. Pioneered a morphological approach, achieving **0.8029 and 0.8988 AUC** scores on both datasets which save thousands.
- The research was **published and presented** at the International Conference on Innovation in Artificial Intelligence in Japan, the published paper can be [viewed](#).

JuanScan: ID Parser Google ML Kit for Android — [Android](#), [Google ML Kit](#), [Python FastAPI](#), [Java](#)

- An Android Application that preprocesses and parses local PH Government IDs, using ML Kit for the OCR.
- Reducing the manual process of writing IDs from **45 seconds down to only 10 seconds**. With full logging features and preprocessing.

Vista Suite: Salon Management — [React](#), [TailwindCSS](#), [Supabase](#), [Express](#), [NodeJS](#)

- Developed a centralized salon management platform that replaced manual handwritten transaction and customer records with a streamlined three-step digital workflow for logging, tracking, and profiling customers.
- Reduced transaction recording time by approximately 50% while eliminating scalability issues associated with paper-based processes.
- Consolidated customer profiles and transaction history into a single database, improving data accuracy, traceability, and accessibility.

Mazvantage: Parts Catalogue Store — [React](#), [TailwindCSS](#), [Supabase](#), [Javascript](#), [SEO](#), [Express](#), [NodeJS](#)

- A catalogue website that sells car accessories and parts for Mazda vehicles. Fully developed within React, along with CRUD operations for the admin, optimized loading times, and equipped with Technical SEO practices.

Twitter/X Affiliate Analytics Dashboard — [Python](#), [OpenAI](#), [Grok](#), [Openclaw](#), [Openrouter](#), [AI](#), [Javascript](#), [NextJS](#), [Express](#), [NodeJS](#)

- A dashboard filled with competitor analysis, post suggestions, and topics, all connected to a main account to provide the user with specialized information regarding influencer activities. A single dashboard for everything an influencer might need saving **manhours**.

Twitter/X Changes Monitor — [Python](#), [Requests](#), [API](#), [Streaming](#)

- Built an automation tool that tracks real-time Twitter/X account activity (bio, profile, and tag updates) and updates via webhook, improving data responsiveness by integrating directly with the API, significantly reducing latency versus manual monitoring.

WORK EXPERIENCE

Lead Developer, Speranza Innovations – Texas, USA

Dec 2025 – Mar 2026

- Built and launched a multi-tenant SaaS platform **serving 2+ companies and individual statewide**, improving operational efficiency by **25–40%** and reducing communication turnaround time by **30–50%**.
- Led end-to-end product engineering (backend, frontend, system architecture, and UI/UX), delivering the platform from zero to production in 3-4 months and shipping **10+ core features** for reporting, collaboration, and workflow automation.
- Architected and implemented a Google Cloud Pub/Sub + catch-all email ingestion pipeline with filtering/routing logic, automating processing of 100+ inbound messages/week and reducing manual effort by **reducing a 10-minute process to only a minute** task.

Software Developer, Lightweight Solutions – Quezon City, Commonwealth

Aug 2025 – Nov 2025

- Led the development of a full-stack web platform for a major local donation and charity organization, integrating payment gateways, monthly subscriptions, and responsive UI/UX designs using modern web technologies. From a previous manual process that takes **3-5 minutes** to complete, **now down to a 1-2 minute** completion time from user information to a convenient checkout page.
- Chatbot engineering to optimize responses, reduce overhead costs,
- Conceptualized and developed an AI-driven SaaS platform that enables users to create and manage custom AI assistants for various business applications. Designed autonomous agentic models to troubleshoot, re-correct, and solve with minimal manual manipulation.

Data Science Intern, Arthaland Corporation – Taguig, BGC

Feb 2024 – May 2024

- Built a data integration tool to merge datasets from multiple database systems, automating a manual process and increasing accuracy and efficiency, achieving over **87.5% daily manual time improvement** that's over **8 manhours saved** per week.
- Leveraged Power BI dashboards to be presented to investors and stakeholders with proper data analysis on where to focus and where not to.

Software Developer, MinTech – Netherlands

Nov 2022 – Aug 2023

- Collaborated with a software development team to design, implement, and maintain scalable solutions, ensuring quality and reliability.
- Developed and deployed a trading bot that enabled users to track trades with real-time statistical insights, increasing time efficiency by **0.5s – 5s** which is critical in the trading scene. Integrated real-time dashboards to track statistics and gains throughout the day.

EDUCATION

University of Santo Tomas - Manila, Philippines

2020-2024

B.S. Computer Science specializing in Data Science; Consistent Dean's Lister, Cum Laude

SKILLS

Programming & Frameworks: Python, JavaScript, TypeScript, Java, Node.js, Express.js, Next.js, React, PostgreSQL, Supabase, Neon

AI & Machine Learning: TensorFlow, Scikit-learn, AI Prompt Engineering, Model Fine-Tuning, Natural Language Processing

Data Science & Analytics: Power BI, Data Visualization, Data Cleaning, Predictive Modeling, Statistical Analysis

Web & Software Development: Full-Stack Development, SaaS Development/Architecture, Agile/Scrum, Responsive Web Design